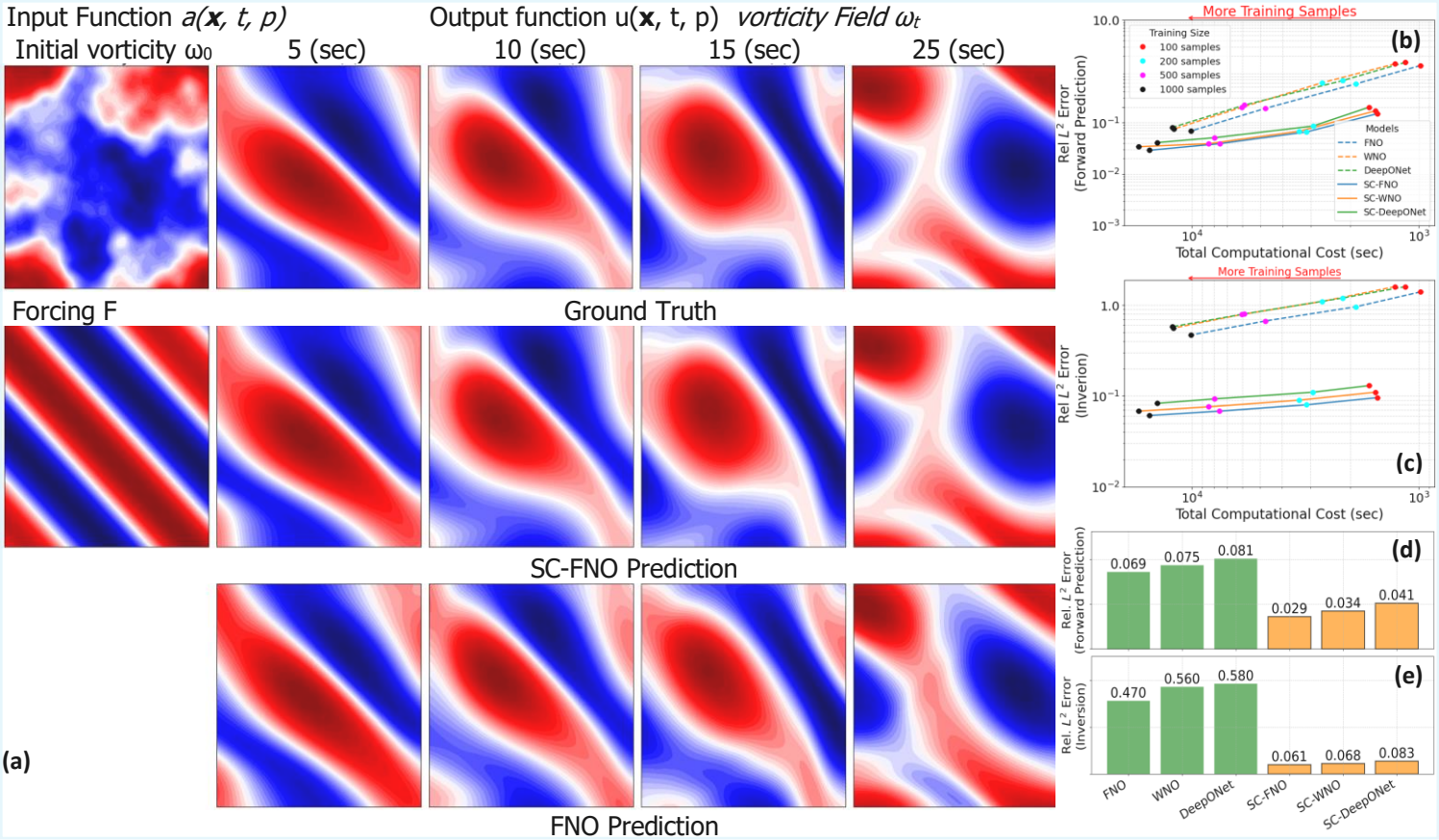
Input Function a(x, t, p)
Output function u(x, t, p) vorticity Field ωt
Initial vorticity ω0
5 (sec)
10 (sec)
15 (sec)
25 (sec)
Forcing F
Ground Truth
SC-FNO Prediction
FNO Prediction
(a)
More Training Samples
Training Size
100 samples
200 samples
500 samples
1000 samples
Models
FNO
WNO
DeepONet
SC-FNO
SC-WNO
SC-DeepONet
(b)
Rel L2 Error (Forward Prediction)
Total Computational Cost (sec)
More Training Samples
(c)
Rel L2 Error (Inverion)
Total Computational Cost (sec)
(d)
Rel. L2 Error (Forward Prediction)
0.069
0.075
0.081
0.029
0.034
0.041
(e)
Rel. L2 Error (Inversion)
0.470
0.560
0.580
0.061
0.068
0.083
FNO
WNO
DeepONet
SC-FNO
SC-WNO
SC-DeepONet